

Within-participant change in reaction time between two assessments

Abstract

Background: A laboratory study of caffeine and attention. Methods: The change in reaction time (ms) on the Stroop task from the first to the second assessment was tested with a paired t-test (two-sided). Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a laboratory study of caffeine and attention. The primary question concerns caffeine administration and reaction time (ms) on the Stroop task. The accompanying dataset (data.csv) contains one row per participant-record as exported from the study database, including some duplicated exports and sentinel-coded missing values; it is described in the data dictionary.

Methods

Participants and data. The export contains 246 records. Variables are listed in the data dictionary. Group membership is recorded in `condition`.

Data cleaning and exclusions. Exact duplicate records (identical on every column) were removed, keeping the first occurrence. The value -999 denotes a missing measurement and was treated as missing. Only participants aged 18 to 80 years inclusive were analysed. Records with missing `rt_ms`, `rt_ms_post` were excluded (complete-case analysis).

Statistical analysis. The change in reaction time (ms) on the Stroop task from the first to the second assessment was tested with a paired t-test (two-sided). The mean change is the mean of (second minus first) over analysed participants; its standard deviation is the sample SD of the paired differences. Cohen's d_z is the mean change divided by the SD of the differences. Percent change was computed per participant as $(\text{second} - \text{first}) / \text{first} \times 100$ and then averaged. All analyses used the cleaned analysis sample; no random resampling was used.

Results

Among 211 participants the mean value changed from 696.07 to 672.71, a mean change of -23.36 (SD 44.46; statistic = -7.63, $p < 0.001$; $d_z = -0.525$); the mean percent change was -3.38%.

Discussion

The analysis followed the pre-specified plan. Limitations include reliance on database exports and complete-case handling of missing values.